

DBMS –SELLER AND INVENTORY MANAGEMENT

LENSKART-E-Commerce Order Management Database System

1.DATABASE DESIGN:

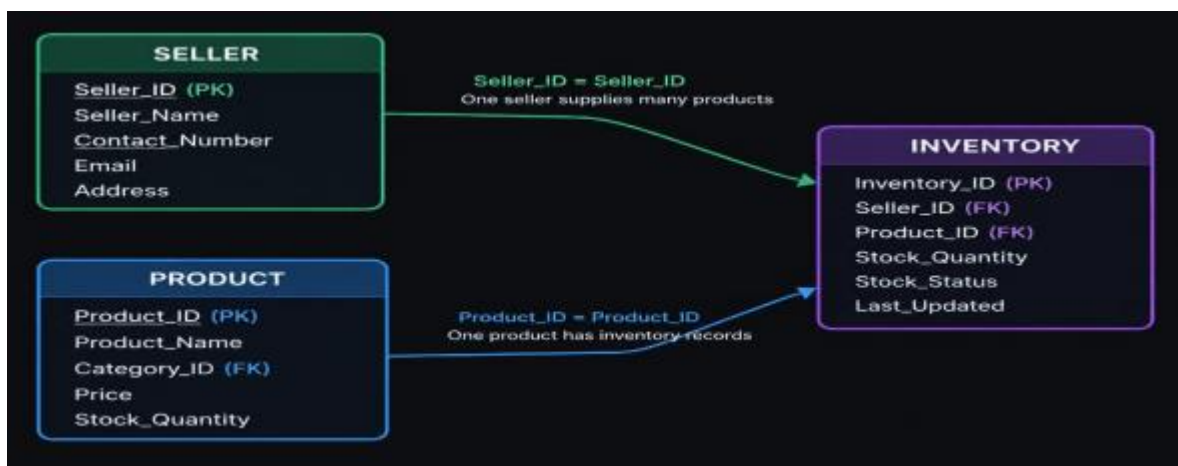
1.1 Seller table :

Attribute	Description
<u>Seller_ID</u>	Unique seller ID
<u>Seller_Name</u>	Seller name
<u>Contact_Number</u>	Seller contact number
<u>Email</u>	Seller email address
<u>Address</u>	Seller address

1.2 Inventory Table :

Attribute	Description
<u>Inventory_ID</u>	Unique inventory ID
<u>Product_ID</u>	Product ID
<u>Seller_ID</u>	Seller ID
<u>Stock_Quantity</u>	Available stock quantity
<u>Stock_Status</u>	Availability status of the product
<u>Last_Updated</u>	Date when inventory was last updated

2. TABLE RELATIONSHIP



3. Table creation

3.1 Seller table :

```
SQL> CREATE TABLE LenskartSeller (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(100) NOT NULL,  
4     Contact_Number VARCHAR2(15),  
5     Email VARCHAR2(100) UNIQUE,  
6     Address VARCHAR2(200)  
7 );
```

Table created.

3.2 Inventory Table :

```
SQL> CREATE TABLE LenskartInventory (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Product_ID NUMBER REFERENCES Product(Product_ID),  
4     Seller_ID NUMBER REFERENCES LenskartSeller(Seller_ID),  
5     Stock_Quantity NUMBER NOT NULL,  
6     Stock_Status VARCHAR2(20) NOT NULL,  
7     Last_Updated DATE NOT NULL  
8 );
```

Table created.

4. Seller data insertion

```
SQL> INSERT INTO LenskartSeller  
2 VALUES (1, 'Vision Plus', '9876543210', 'visionplus@gmail.com', 'Chennai');
```

1 row created.

```
SQL>  
SQL> INSERT INTO LenskartSeller  
2 VALUES (2, 'Eye World', '9876543211', 'eyeworld@gmail.com', 'Bangalore');
```

1 row created.

```
SQL>  
SQL> INSERT INTO LenskartSeller  
2 VALUES (3, 'Optic Zone', '9876543212', 'opticzone@gmail.com', 'Coimbatore');
```

1 row created.

```
SQL>  
SQL> INSERT INTO LenskartSeller  
2 VALUES (4, 'Clear Vision Store', '9876543213', 'clearvision@gmail.com', 'Madurai');
```

1 row created.

```
SQL>  
SQL> INSERT INTO LenskartSeller  
2 VALUES (5, 'Lens World', '9876543214', 'lensworld@gmail.com', 'Salem');
```

5. Inventory data insertion :

```
SQL> INSERT INTO LenskartInventory
  2  VALUES (1, 1, 1, 50, 'Available', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO LenskartInventory
  2  VALUES (2, 2, 2, 30, 'Available', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO LenskartInventory
  2  VALUES (3, 3, 3, 25, 'Available', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO LenskartInventory
  2  VALUES (4, 4, 4, 40, 'Available', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO LenskartInventory
  2  VALUES (5, 5, 5, 0, 'Unavailable', SYSDATE);
```

6. SELLER PRODUCT INFORMATION

6.1 Seller and Product Details

```
SQL> SELECT
  2     S.Seller_ID,
  3     S.Seller_Name,
  4     P.Product_ID,
  5     P.Product_Name,
  6     I.Stock_Quantity
  7 FROM Seller S
  8 JOIN Inventory I
  9     ON S.Seller_ID = I.Seller_ID
 10 JOIN Product P
 11     ON I.Product_ID = P.Product_ID;

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
          1
Vision Plus
          1

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
Classic Black Frame
          50
```

```
SQL Plus

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
      2
Eye World
      2

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
Blue Light Glasses
      30

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
      3
```

SELLER_ID	
SELLER_NAME	
PRODUCT_ID	
PRODUCT_NAME	
STOCK_QUANTITY	
	3
Optic Zone	3
SELLER_ID	
SELLER_NAME	
PRODUCT_ID	
PRODUCT_NAME	
STOCK_QUANTITY	
Round Metal Frame	0
SELLER_ID	
SELLER_NAME	
PRODUCT_ID	
PRODUCT_NAME	
STOCK_QUANTITY	
	1
Vision Plus	4

```
SQL Plus
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
Classic Black Sunglasses
      25

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
      2
Eye World
      5

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY
-----
Daily Contact Lenses
      60
```

```
SQL Plus
PRODUCT_NAME
-----
BRAND
-----
      PRICE STOCK_QUANTITY
-----
Optic Zone
Round Metal Frame
Lenskart Air
      1500           0

SELLER_NAME
-----
PRODUCT_NAME
-----
BRAND
-----
      PRICE STOCK_QUANTITY
-----
Vision Plus
Classic Black Sunglasses
Fastrack
      2000           25

SELLER_NAME
-----
PRODUCT_NAME
-----
BRAND
-----
      PRICE STOCK_QUANTITY
-----
Eye World
Daily Contact Lenses
Bausch 10000
      900           60

SQL>
```

7. INVENTORY STATUS

7.1 Display Available Products

```
SQL Plus
SQL>
SQL> SELECT
2     I.Inventory_ID,
3     P.Product_Name,
4     I.Stock_Quantity,
5     CASE
6         WHEN I.Stock_Quantity > 0 THEN 'Available'
7         ELSE 'Unavailable'
8     END AS Product_Status
9 FROM Inventory I
10 JOIN Product P
11     ON I.Product_ID = P.Product_ID;

INVENTORY_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STA
-----
          1
Classic Black Frame
          50 Available

          2
Blue Light Glasses
          30 Available

INVENTORY_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STA
-----

          3
Round Metal Frame
          0 Unavailable

          4
Classic Black Sunglasses
```

```
SQL Plus

INVENTORY_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STA
-----
          25 Available

          5
Daily Contact Lenses
          60 Available
```

7.2 Display Unavailable Products

```
SQL> SELECT
 2     I.Inventory_ID,
 3     P.Product_Name,
 4     S.Seller_Name,
 5     I.Stock_Quantity,
 6     I.Reorder_Level,
 7     CASE
 8         WHEN I.Stock_Quantity = 0 THEN 'Unavailable'
 9         WHEN I.Stock_Quantity <= I.Reorder_Level THEN 'Low Stock'
10         ELSE 'Available'
11     END AS Inventory_Status
12 FROM Inventory I
13 JOIN Product P
14     ON I.Product_ID = P.Product_ID
15 JOIN Seller S
16     ON I.Seller_ID = S.Seller_ID;
```

```
INVENTORY_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY REORDER_LEVEL INVENTORY_S
-----
1
Classic Black Frame
```

```
SQL Plus
Classic Black Frame
Vision Plus
50          10 Available

INVENTORY_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY REORDER_LEVEL INVENTORY_S
-----
2
Blue Light Glasses
Eye World
30          5 Available
```

```
INVENTORY_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY REORDER_LEVEL INVENTORY_S
-----
3
Round Metal Frame
Optic Zone
0           10 Unavailable

INVENTORY_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY REORDER_LEVEL INVENTORY_S
```

```

SQL Plus
x + v
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY REORDER_LEVEL INVENTORY_S
-----
3
Round Metal Frame
Optic Zone
0 10 Unavailable

INVENTORY_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY REORDER_LEVEL INVENTORY_S
-----
4
Classic Black Sunglasses
Vision Plus
25 5 Available

INVENTORY_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY REORDER_LEVEL INVENTORY_S
-----
5
Daily Contact Lenses
Eye World
60 10 Available

SQL>

```

8. Inventory Status

8.1 Available and Unavailable Product Count

```

SQL> SELECT
2     Stock_Status,
3     COUNT(*) AS Total_Products
4 FROM LenskartInventory
5 GROUP BY Stock_Status
6 ORDER BY Stock_Status;

```

```

STOCK_STATUS      TOTAL_PRODUCTS
-----
Available          4
Unavailable         1

```

8.2 Inventory Summary :

```
SQL> SELECT
  2     COUNT(*) AS Total_Inventory,
  3     SUM(CASE
  4         WHEN Stock_Status = 'Available' THEN 1
  5         ELSE 0
  6     END) AS Available_Products,
  7     SUM(CASE
  8         WHEN Stock_Status = 'Unavailable' THEN 1
  9         ELSE 0
 10     END) AS Unavailable_Products,
 11     SUM(Stock_Quantity) AS Total_Stock
 12 FROM LenskartInventory;
```

TOTAL_INVENTORY	AVAILABLE_PRODUCTS	UNAVAILABLE_PRODUCTS	TOTAL_STOCK
5	4	1	145